

Midterm CSCI6032 Machine Learning — Answers

Fall 2026

Worked solutions for all seven problems. Problems 1–6 total 100 points; Problem 7 is worth 10 extra-credit points.

Problem 1 (18 points total)

Given the matrices:

$$A = \begin{bmatrix} 2 & 3 \\ 5 & 4 \\ -1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 1 \\ 2 & 3 \\ 2 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix}$$

(a) Compute $A^T B - C$. (5 points)

Answer. First multiply:

$$\begin{aligned} A^T B &= \begin{bmatrix} 2 & 5 & -1 \\ 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} 5 & 1 \\ 2 & 3 \\ 2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2(5) + 5(2) + (-1)(2) & 2(1) + 5(3) + (-1)(0) \\ 3(5) + 4(2) + 2(2) & 3(1) + 4(3) + 2(0) \end{bmatrix} \\ &= \begin{bmatrix} 18 & 17 \\ 27 & 15 \end{bmatrix}. \end{aligned}$$

Then subtract:

$$A^T B - C = \begin{bmatrix} 18 - 4 & 17 - 2 \\ 27 - 0 & 15 - 1 \end{bmatrix} = \begin{bmatrix} 14 & 15 \\ 27 & 14 \end{bmatrix}.$$

(b) Compute $3C^{-1}C$. (5 points)

Answer. Since $\det(C) = 4 \neq 0$, the matrix is invertible and $C^{-1}C = I_2$. Therefore,

$$3C^{-1}C = 3I_2 = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}.$$

An explicit inverse, if computed, is

$$C^{-1} = \begin{bmatrix} \frac{1}{4} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix}.$$

(c) The Hadamard (element-wise) product is denoted $\odot$. Compute $(A \odot B)C$ (8 points)

Answer. First multiply corresponding entries:

$$A \odot B = \begin{bmatrix} 2(5) & 3(1) \\ 5(2) & 4(3) \\ (-1)(2) & 2(0) \end{bmatrix} = \begin{bmatrix} 10 & 3 \\ 10 & 12 \\ -2 & 0 \end{bmatrix}.$$

Then use ordinary matrix multiplication:

$$(A \odot B)C = \begin{bmatrix} 10 & 3 \\ 10 & 12 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 40 & 23 \\ 40 & 32 \\ -8 & -4 \end{bmatrix}.$$

Problem 2 (18 points total)

Given the column vectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -3 \\ 6 \end{bmatrix}$$

- (a) Compute the dot product $\mathbf{u} \cdot \mathbf{v}$, i.e., $\mathbf{u}^T \mathbf{v}$. (5 points)

Answer.

$$\mathbf{u}^T \mathbf{v} = 1(3) + 4(-3) + 2(6) = 3 - 12 + 12 = 3.$$

- (b) Compute the outer product $\mathbf{u}\mathbf{v}^T$. (5 points)

Answer.

$$\mathbf{u}\mathbf{v}^T = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & -3 & 6 \end{bmatrix} = \begin{bmatrix} 3 & -3 & 6 \\ 12 & -12 & 24 \\ 6 & -6 & 12 \end{bmatrix}.$$

- (c) Given the column vector

$$\mathbf{w} = \begin{bmatrix} 13 \\ 7 \\ 26 \end{bmatrix}$$

Express $\mathbf{w}$ as a linear combination of $\mathbf{u}$ and $\mathbf{v}$. (8 points)

Answer. Seek a and b such that $a\mathbf{u} + b\mathbf{v} = \mathbf{w}$. Matching coordinates gives

$$\begin{aligned} a + 3b &= 13, \\ 4a - 3b &= 7, \\ 2a + 6b &= 26. \end{aligned}$$

Adding the first two equations gives $5a = 20$, so $a = 4$. Substituting into the first gives $b = 3$. The third equation also holds: $2(4) + 6(3) = 26$. Thus,

$$\mathbf{w} = 4\mathbf{u} + 3\mathbf{v}.$$

Problem 3 (12 points total)

Column vectors are *linearly independent* if none can be formed as a linear combination of the others.

Given the matrix:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 6 & 12 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- (a) How many linearly independent columns are there? Can any be expressed as a linear combination of the others? If yes, which ones? (6 points)

Answer. Denote the columns by

$$c_1 = \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 2 \\ 12 \\ 0 \end{bmatrix}, \quad c_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Because $c_2 = 2c_1$, all three columns together are linearly dependent. However, c_1 and c_3 are linearly independent: if $\alpha c_1 + \beta c_3 = \mathbf{0}$, the first and third coordinates give $\alpha = \beta = 0$. The maximum number of linearly independent columns is therefore

$$\boxed{2.}$$

One independent set is $\{c_1, c_3\}$; another is $\{c_2, c_3\}$. Both of the first two columns can be expressed using the others:

$$\boxed{c_2 = 2c_1 + 0c_3, \quad c_1 = \frac{1}{2}c_2 + 0c_3.}$$

The third column cannot be expressed using the first two, because every linear combination of those columns has third coordinate zero.

(b) What is the rank of matrix A ? (3 points)

Answer. The rank is the maximum number of linearly independent columns:

$$\boxed{\text{rank}(A) = 2.}$$

(c) What is the span of the columns of matrix A ? (3 points)

Answer. Since $c_2 = 2c_1$, the second column does not enlarge the span:

$$\begin{aligned} \text{Col}(A) &= \text{span}\{c_1, c_2, c_3\} \\ &= \text{span}\left\{\begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\} \\ &= \left\{s \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} : s, t \in \mathbb{R}\right\}. \end{aligned}$$

Equivalently,

$$\boxed{\text{Col}(A) = \left\{\begin{bmatrix} s \\ 6s \\ t \end{bmatrix} : s, t \in \mathbb{R}\right\}.$$

Geometrically, this is the plane $z_2 = 6z_1$ in $\mathbb{R}^3$, where (z_1, z_2, z_3) denotes a point in the plane.

Problem 4 (18 points total)

Yanhui Gang Electric manufactures two types of light bulbs: **Type A** and **Type B**. It is known that:

- 30% of the light bulbs produced are **Type A**, and 70% are **Type B**.
- 2% of **Type A** light bulbs are defective, while 5% of **Type B** light bulbs are defective.

Joe Customer buys a light bulb at random and discovers that it is defective. Let D represent the event that a bulb is defective, and let A and B represent the events that the bulb is of **Type A** and **Type B**, respectively.

(a) What is the probability that a randomly selected light bulb is defective? (9 points)

Answer. The events A and B partition the possible bulb types. By the law of total probability,

$$\begin{aligned} P(D) &= P(D | A)P(A) + P(D | B)P(B) \\ &= (0.02)(0.30) + (0.05)(0.70) \\ &= 0.006 + 0.035 = 0.041. \end{aligned}$$

Thus,

$$\boxed{P(D) = 0.041 = 4.1\%.$$

- (b) Given that the light bulb is defective, what is the probability that it is a **Type A** bulb? Use **Bayes' law**. (9 points)

Answer. By Bayes' law,

$$\begin{aligned} P(A | D) &= \frac{P(D | A)P(A)}{P(D)} \\ &= \frac{(0.02)(0.30)}{0.041} \\ &= \frac{0.006}{0.041} = \frac{6}{41}. \end{aligned}$$

Therefore,

$$P(A | D) = \frac{6}{41} \approx 0.14634 \approx 14.63\%.$$

Problem 5 (18 points total)

- (a) Come up with an example of two discrete random variables X and Y where $E[XY] \neq E[X]E[Y]$. Provide a probability assignment as a table with X outcomes along the side and Y outcomes along the top wherein each entry represents the probability that $X = x$ and $Y = y$. Show how this demonstrates that $E[XY] \neq E[X]E[Y]$ (9 points)

Answer. One valid example is the following joint distribution:

$X \backslash Y$	$Y = 0$	$Y = 1$
$X = 0$	$\frac{1}{2}$	0
$X = 1$	0	$\frac{1}{2}$

All entries are nonnegative and sum to 1. Here X and Y always take the same value, with 0 and 1 each occurring with probability $\frac{1}{2}$. The marginal distributions give

$$E[X] = 0 \left(\frac{1}{2} \right) + 1 \left(\frac{1}{2} \right) = \frac{1}{2}, \quad E[Y] = \frac{1}{2}.$$

Using the joint distribution,

$$\begin{aligned} E[XY] &= \sum_x \sum_y xyP(X = x, Y = y) \\ &= (0)(0) \left(\frac{1}{2} \right) + (0)(1)(0) + (1)(0)(0) + (1)(1) \left(\frac{1}{2} \right) \\ &= \frac{1}{2}. \end{aligned}$$

Consequently,

$$E[XY] = \frac{1}{2} \neq \frac{1}{4} = E[X]E[Y].$$

Other valid joint distributions satisfying this inequality are also acceptable.

- (b) Show that if two discrete random variables X and Y are independent, then their covariance is zero. Show this mathematically starting from the definitions of independence, expectation, and covariance. (9 points)

- Independence means that for all outcomes of X and Y : $P(X = x, Y = y) = P(X = x)P(Y = y)$
- Expectation is $E[X] = \sum_x xP(X = x)$
- $\text{Cov}[X, Y] = E[(X - E[X])(Y - E[Y])]$.

- $E[XY] = \sum_x \sum_y xyP(X = x, Y = y)$

Answer. Assume the expectations involved are finite; this is automatic when X and Y each take finitely many values.

Independence means

$$P(X = x, Y = y) = P(X = x)P(Y = y).$$

Substituting into the definition of $E[XY]$ and factoring the sums gives

$$\begin{aligned} E[XY] &= \sum_x \sum_y xyP(X = x, Y = y) \\ &= \sum_x \sum_y xyP(X = x)P(Y = y) \\ &= \left(\sum_x xP(X = x) \right) \left(\sum_y yP(Y = y) \right) \\ &= E[X]E[Y]. \end{aligned}$$

Now start from the definition of covariance and expand. Let $\mu_X = E[X]$ and $\mu_Y = E[Y]$, which are constants:

$$\begin{aligned} \text{Cov}[X, Y] &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y] \\ &= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y \\ &= E[XY] - E[X]E[Y]. \end{aligned}$$

Using the identity just proved,

$$\text{Cov}[X, Y] = E[X]E[Y] - E[X]E[Y] = 0.$$

Problem 6 (16 points total)

We fit a linear regression model with design matrix $X \in \mathbb{R}^{m \times (n+1)}$, where m is the number of observations and n is the number of features. The first column of X is all ones so that the intercept term θ_0 can be included in Θ . The parameter vector is $\Theta \in \mathbb{R}^{n+1}$. Let $y \in \mathbb{R}^m$.

The cost function is the mean squared error:

$$J(\Theta) = \frac{1}{m} \sum_{i=1}^m (y^{(i)} - x^{(i)}\Theta)^2$$

where $x^{(i)}$ is the i -th row of X .

Show that this cost function can be written in matrix notation as $J(\Theta) = \frac{1}{m} \|y - X\Theta\|^2$, where $\|v\|^2 = v^T v$. (16 points)

Answer. Each $x^{(i)}$ is a row vector with $n + 1$ entries, and Θ is a column vector with $n + 1$ entries. Thus $x^{(i)}\Theta$ is the scalar prediction for observation i .

Stacking the rows and predictions gives

$$X = \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(m)} \end{bmatrix}, \quad X\Theta = \begin{bmatrix} x^{(1)}\Theta \\ x^{(2)}\Theta \\ \vdots \\ x^{(m)}\Theta \end{bmatrix}.$$

The residual vector at an arbitrary Θ is therefore

$$e = y - X\Theta = \begin{bmatrix} y^{(1)} - x^{(1)}\Theta \\ y^{(2)} - x^{(2)}\Theta \\ \vdots \\ y^{(m)} - x^{(m)}\Theta \end{bmatrix}.$$

Its squared Euclidean norm is the sum of its squared entries:

$$\begin{aligned} \|y - X\Theta\|^2 &= (y - X\Theta)^T (y - X\Theta) \\ &= e^T e \\ &= \sum_{i=1}^m (y^{(i)} - x^{(i)}\Theta)^2. \end{aligned}$$

Dividing by m gives

$$J(\Theta) = \frac{1}{m} \|y - X\Theta\|^2 = \frac{1}{m} (y - X\Theta)^T (y - X\Theta).$$

Problem 7 (10 points extra credit for all students)

This problem is extra credit for everyone.

This problem continues the setup from Problem 6: X , Θ , $J(\Theta)$, and y are as defined there.

Let the residual vector be $r = y - X\hat{\Theta}$, where $\hat{\Theta}$ minimizes $J(\Theta)$. Show that r is orthogonal to each column of X , meaning $X^T r = \mathbf{0}$.

Note: As derived in class, any minimizer $\hat{\Theta}$ satisfies the normal equations $X^T X \hat{\Theta} = X^T y$. If $X^T X$ is invertible, these equations give $\hat{\Theta} = (X^T X)^{-1} X^T y$. (10 points)

Answer. Because $\hat{\Theta}$ minimizes the squared-error cost, it satisfies the normal equations

$$X^T X \hat{\Theta} = X^T y.$$

Substituting $r = y - X\hat{\Theta}$ gives

$$\begin{aligned} X^T r &= X^T (y - X\hat{\Theta}) \\ &= X^T y - X^T X \hat{\Theta} \\ &= X^T y - X^T y \\ &= \mathbf{0}. \end{aligned}$$

To interpret this column by column, write $X = [c_0 \ c_1 \ \cdots \ c_n]$. Then

$$X^T r = \begin{bmatrix} c_0^T r \\ c_1^T r \\ \vdots \\ c_n^T r \end{bmatrix} = \mathbf{0}.$$

Each entry is a dot product, so

$$c_j^T r = 0 \quad \text{for every } j = 0, \dots, n.$$

Hence r is orthogonal to every column of X . This argument does not require $X^T X$ to be invertible.